Problem Statement

Automated Gravity-Fed Object Counting, Sizing, and Jam Detection System

In a baggage-handling line, detecting an item at the entry does not confirm that it has reached the exit. An item may enter a monitored section and become stuck between the two points. If this delay goes unnoticed, following items may accumulate, interrupting the flow and increasing the risk of damage and handling delays. A basic counter at a single location cannot establish whether an entered item has completed its passage, while manual observation requires an operator to watch the section continuously.

The central problem is therefore to determine whether each object entering a monitored transport section reaches the exit within an allowed time, maintain a meaningful passage count, and notify the operator when an object remains inside too long. After the obstruction is removed, the system should register the delayed arrival and clear the jam alert when no objects remain in the section. It should also flag prolonged absence of completed passages so that the operator can investigate inactivity.

Counting alone provides limited information about the movement being monitored. The operator also needs an estimate of object speed and length, a basic size category, and an indication of unexpected exit-side detections that may represent reverse movement. These observations help explain the condition of the section. They should be available through local indicators and a wireless terminal, reducing the need for continuous visual supervision.

This project addresses these needs through a low-cost laboratory prototype built around an STM32F401CCU6 microcontroller programmed in bare-metal C. Two TCRT5000 infrared sensors, placed 150 mm apart on a gravity-fed chute, provide entry and exit events. Their timing is used to count forward passages and estimate transit speed; the duration of detection at the exit sensor is combined with that speed to estimate object length and assign small, standard, or oversized prototype categories. The controller tracks pending arrivals, detects excessive transit delay, provides buzzer and LED indications, transmits status through HC-05 Bluetooth, and supports a manual reset.

The current firmware uses a fixed two-second transit limit and a twenty-second inactivity limit after a completed passage. The chute and test objects model a baggage-conveyor section for one-object-at-a-time demonstrations. Length is an estimate, and inactivity alone does not establish that a conveyor motor has stopped. The project demonstrates monitoring and operator alerts; automatic belt stopping, physical sorting, full luggage dimension measurement, and validation on an industrial conveyor are outside its present scope.